

# gemstone-py vs gemstone-rs

Install paths, use cases, maturity, and feature matrix



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# gemstone-py vs gemstone-rs

`gemstone-py` and `gemstone-rs` solve related problems at different maturity levels.

Use `gemstone-py` when you want the most complete Python experience today: Python sessions, async examples, web framework adapters, native acceleration, codegen, VS Code workbench integration, and a Python database explorer.

Use `gemstone-rs` when you want Rust services, CLIs, workers, or tooling to talk to GemStone/S directly without keeping Python in the process.

## Install Paths

| Goal                  | gemstone-py                                                                                                          | gemstone-rs                                                                                                                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Application library   | <code>python -m pip install gemstone-py</code>                                                                       | <code>cargo add gemstone-rs</code>                                                                                                                                                                                       |
| Native acceleration   | <code>python -m pip install "gemstone-py[fast]"</code>                                                               | Built into the Rust GCI path once <code>libgcirpc</code> is available                                                                                                                                                    |
| Examples from source  | <code>python -m pip install -e ".[examples]"</code>                                                                  | <code>cargo run -p gemstone-rs --example quickstart</code>                                                                                                                                                               |
| Example discovery/run | <code>gemstone-examples hello</code> , <code>list</code> , <code>plan3-map</code> , and <code>launch</code> commands | <code>gemstone-rs hello</code> , <code>compare gemstone-py</code> , <code>examples list</code> , <code>map</code> , <code>show</code> , <code>source-checkout run</code> , and installed <code>scaffold</code> templates |
| CLI tooling           | Python modules and examples                                                                                          | <code>cargo install gemstone-rs-cli</code>                                                                                                                                                                               |
| Local explorer        | <code>python-gemstone-database-explorer</code>                                                                       | <code>cargo install gemstone-rs-explorer</code>                                                                                                                                                                          |
| VS Code               | <code>gemstone-py</code> Workbench                                                                                   | <code>gemstone-rs</code> Workbench                                                                                                                                                                                       |

Both projects use the same GemStone-style environment:

```
export GS_LIB=/opt/gemstone/product/lib
export GS_STONE=gs64stone
export GS_USERNAME=DataCurator
export GS_PASSWORD=your-password
```

`GS_STONE_NAME` is accepted as a stone-name alias where the clients support it. Use `GS_LIB_PATH` when you want to point directly at a specific `libgcirpc` file.

## Best Fit

| Use case                          | Better choice                              | Why                                                                                                                                                                                                               |
|-----------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Python web app                    | gemstone-py                                | FastAPI and Litestar examples are already first-class.                                                                                                                                                            |
| Python scripts and notebooks      | gemstone-py                                | Lower friction for Python teams.                                                                                                                                                                                  |
| Rust service or worker            | gemstone-rs                                | Native Rust API, Rust errors, Rust ownership, Cargo distribution.                                                                                                                                                 |
| Rust CLI tooling                  | gemstone-rs                                | CLI and explorer can run without Python.                                                                                                                                                                          |
| Example-driven onboarding         | gemstone-py today, gemstone-rs catching up | gemstone-py has broader runnable examples; gemstone-rs now has no-live hello, a direct comparison command, an installed CLI example index, feature map, source-checkout runner, and standalone project scaffolds. |
| VS Code database/codegen workflow | gemstone-py today, gemstone-rs later       | gemstone-py has the more mature explorer; gemstone-rs has the cleaner Rust backend direction.                                                                                                                     |
| Shared low-level bridge           | gemstone-rs                                | The Rust API is the right place to isolate GCI safety and ownership.                                                                                                                                              |

## Maturity Matrix

| Capability          | gemstone-py               | gemstone-rs                                                                                                         |
|---------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------|
| Stable install path | Mature: PyPI and TestPyPI | Early: crates.io workflow and verification added                                                                    |
| Native bridge       | PyO3 extension path       | Rust GCI crate and safe API                                                                                         |
| Sync API            | Mature                    | Initial safe API                                                                                                    |
| Async API           |                           | Dedicated-thread <code>SessionWorker</code> and bounded <code>SessionWorkerPool</code> ; async facade still planned |

| Capability         | gemstone-py                          | gemstone-rs                                                                                                                                                                                                                    |
|--------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | Mature enough for examples and tests |                                                                                                                                                                                                                                |
| Web frameworks     | FastAPI, Litestar, Django examples   | Shared <code>gemstone_rs::web</code> health helpers, standard-library HTTP, <code>SessionWorkerPool</code> , packaged <code>gemstone-rs-axum</code> / <code>gemstone-rs-actix</code> adapters, and checked Axum/Actix examples |
| Codegen            | Python wrapper workflow              | Rust wrapper workflow with preview/diff/check/generate, live discovery, and profile scaffolds                                                                                                                                  |
| Browser API        | Used by database explorer            | CLI/explorer API for dictionaries/classes/methods/source                                                                                                                                                                       |
| Local explorer     | More mature Python app               | Minimal Rust explorer proving the API                                                                                                                                                                                          |
| VS Code extension  | More complete product flow           | Thin command/workbench layer over Rust CLI and explorer                                                                                                                                                                        |
| Live tests         | Broader coverage                     | Growing smoke suite for login/eval/perform/globals/transactions/codegen                                                                                                                                                        |
| Release automation | More complete                        | Added crates/VSIX/GitHub verification path                                                                                                                                                                                     |
| Docs               | Broader and polished                 | Catching up with guide, cookbook, article, comparison, example index, PDFs                                                                                                                                                     |

## Feature Matrix

| Feature                       | gemstone-py status | gemstone-rs status |
|-------------------------------|--------------------|--------------------|
| Login/logout                  | Supported          | Supported          |
| <code>3 + 4 == 7</code> smoke | Supported          | Supported          |
| Global put/get                | Supported          | Supported          |
| String round-trip             | Supported          | Supported          |
|                               | Supported          | Supported          |

| Feature                                                 | gemstone-py status                                                                | gemstone-rs status                                                                                                                                                                                                 |
|---------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>perform</code> and <code>perform_oop</code>       |                                                                                   |                                                                                                                                                                                                                    |
| Commit/abort                                            | Supported                                                                         | Supported                                                                                                                                                                                                          |
| Dedicated session worker                                | Supported through Python web integration patterns                                 | Supported through <code>SessionWorker</code> and <code>SessionWorkerPool</code>                                                                                                                                    |
| OOP handles/export set                                  | Supported                                                                         | Supported                                                                                                                                                                                                          |
| Browser dictionaries/ classes/protocols/ methods/source | Supported                                                                         | Supported through <code>Browser</code> , CLI, and explorer                                                                                                                                                         |
| Generated wrappers                                      | Supported                                                                         | Supported for selected classes/methods                                                                                                                                                                             |
| Codegen diff/check/ generate                            | Supported                                                                         | Supported                                                                                                                                                                                                          |
| Live codegen discovery                                  | Supported                                                                         | Supported through CLI/API and installed scaffolds                                                                                                                                                                  |
| FastAPI demo                                            | Supported                                                                         | Not applicable                                                                                                                                                                                                     |
| Litestar demo                                           | Supported                                                                         | Not applicable                                                                                                                                                                                                     |
| Rust HTTP service demo                                  | Not applicable                                                                    | Supported through <code>cargo run -p gemstone-rs --example http_service -- --port 3000</code>                                                                                                                      |
| Axum/Actix demo                                         | Not applicable                                                                    | Axum and Actix services supported through <code>gemstone-rs-axum</code> , <code>gemstone-rs-actix</code> , <code>examples/axum-service</code> , and <code>examples/actix-service</code>                            |
| Installed examples index                                | <code>gemstone-examples hello</code> , <code>list</code> , <code>plan3-map</code> | <code>gemstone-rs hello</code> , <code>compare gemstone-py</code> , <code>examples list</code> , <code>map</code> , <code>show</code> , <code>run --dry-run</code> , <code>scaffold</code> , plus JSON for tooling |
| Database explorer                                       | Mature Python app                                                                 | Rust explorer has browser UI and local API; still less polished                                                                                                                                                    |
| VS Code sidebar workflow                                | More complete                                                                     | Rust workbench has sidebar commands and embedded explorer webview                                                                                                                                                  |

# Actionable Gap Report

The CLI can print the remaining gemstone-py parity gaps directly:

```
gemstone-rs compare gemstone-py --gaps
gemstone-rs compare gemstone-py --gaps --json
```

The report is intentionally action-oriented. Each row names the gemstone-py strength, the gemstone-rs gap, the next implementation step, and the command or test that should verify it.

| Priority | Area                         | What gemstone-py has today                                                            | gemstone-rs next action                                                                                                                                                       |
|----------|------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1       | Web framework adapters       | FastAPI, Litestar, and Django examples are first-class.                               | Add request tracing, middleware examples, and stricter live route smoke coverage around the packaged <code>gemstone-rs-axum</code> and <code>gemstone-rs-actix</code> crates. |
| P1       | Explorer product polish      | The Python database explorer is the richer class browser and product reference.       | Make the embedded Rust explorer webview the primary IDE surface for browsing, codegen, diff, and BridgeRoot inspection.                                                       |
| P1       | Installed example experience | <code>gemstone-examples</code> launches installed examples without a source checkout. | Expand <code>gemstone-rs examples scaffold</code> to explorer-integrated projects and richer generated wrapper profile variants.                                              |
| P2       | Async facade                 | gemstone-py has async examples and FastAPI integration.                               | Add an async facade over <code>SessionWorkerPool</code> after GCI thread behavior is proven with live tests.                                                                  |
| P2       | Shared native core           | gemstone-py already exposes Python packaging and optional native acceleration.        | Make <code>gemstone-py-native</code> a thin PyO3 adapter over <code>gemstone-gci</code> and <code>gemstone-rs</code> .                                                        |
| P2       | Release lane depth           | gemstone-py has mature PyPI/TestPyPI/native wheel/VSIX release lanes.                 | Exercise the full crates.io, Marketplace, GitHub Release, PDF, and checksum workflow regularly.                                                                               |

# Current Gap Analysis

`gemstone-py` remains better when the user wants a batteries-included Python product surface: package extras, web adapters, async examples, a mature database explorer, and an examples launcher aimed at Python users and installed environments.

The most useful `gemstone-rs` catch-up work is to keep copying the product shape, not the implementation language:

- Installed discoverability: `gemstone-rs examples list` now mirrors the `gemstone-py` example map and gives VS Code structured JSON.
- No-live sanity check: `gemstone-rs hello` mirrors `gemstone-examples hello` so users can verify the CLI before configuring GemStone.
- Direct comparison: `gemstone-rs compare gemstone-py` turns this guide into a compact CLI summary with JSON output for tooling.
- Feature stream map: `gemstone-rs examples map` mirrors `gemstone-examples plan3-map`, but frames each stream in Rust crates, examples, docs, and `gemstone-py` reference points.
- Source-checkout launching: `gemstone-rs examples run <name>` now executes the matching Cargo example, with `--dry-run` for CI and documentation checks.
- Installed project scaffolding: `gemstone-rs examples scaffold quickstart`, `browser`, `bridge_root_mapping`, `derive_mapping`, `codegen_preview`, `codegen_workflow`, `codegen_discover`, `codegen_discover_mapping`, `profile_codegen_workflow`, `generated_wrapper_app`, `generated_mapping_app`, `http_service`, `axum_service`, and `actix_service` create standalone Cargo projects from the installed CLI, reducing the need for a source checkout. `profile_codegen_workflow` includes codegen config and project profile files, not only Rust source.
- Rust web service shape: `http_service` gives a real HTTP example with `/`, `/health/local`, and `/health/gemstone` without adding framework dependencies. `gemstone-rs-axum` and `gemstone-rs-actix` now package the same route contract for framework users, and `examples/axum-service` plus `examples/actix-service` are checked crates using those adapters. They can start before credentials are configured and report `/health/gemstone` as a `503` JSON error until the pool is available. They also emit



diagnostic adapter and route headers for route smoke tests and proxy logs. The installed CLI can scaffold `session_worker_pool`, `axum_service`, and `actix_service`. Richer framework middleware and an async facade remain future work.

- Editor workflow: `GemStone RS: Show Example Commands` exposes the same map in the Rust workbench and can run selected Cargo examples in a terminal.
- Remaining gap: `gemstone-rs` still needs installed templates for explorer-integrated workflows and richer generated wrapper profile variants, plus framework middleware, an async facade, and richer explorer screens before its onboarding feels as complete as `gemstone-py`.

## How They Should Work Together

The best long-term shape is not competition between the projects. It is a clean boundary:

- `gemstone-rs` owns the direct Rust/GCI interface and safe Rust API.
- `gemstone-py-native` can become a thin PyO3 layer over the Rust core.
- `gemstone-py` remains the Python API, examples, and Python web integration.
- The Python and Rust explorers can share concepts and eventually converge on common codegen workflows.

That gives Python users the friendliest path and Rust users a direct path, while keeping unsafe GCI behavior isolated in one Rust layer.

## Recommendation

For production Python projects, start with `gemstone-py`.

For Rust-native services, CLIs, workers, and tooling, start with `gemstone-rs`, but treat it as an early API that should be validated against a live stone in your environment.

For VS Code and visual codegen work, use the existing Python explorer as the product reference and keep moving `gemstone-rs-explorer` toward the same level of capability.

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